SIME DARBY PROPERTY · UNIVERSITY HACKATHON 2026

SunD

Progress update

Mentor check-in · 11 Nov 2026 · Team UrbanByte

Transmission cost	Use solar **where it's made**; grid only tops up	p3–4
Upfront cost	**RM 1.76M** per bay · buyers fund the solar	p5–6
SunD e-Panel	**Rejected**	p7
Saver + SunD Pass	Saver **chosen** · Pass for **public drivers**	p8
Business model	NPV **RM 2.26M** · IRR **12%** · payback **9 yrs**	p9–10

Chia Cai Hong · Tae Jie Hui · Chan Kin Chung



Figures: team Google Sheet "SunD financial model and research tracker" and research doc (10 Nov 2026). Details and sources on the pages shown.

RECAP

The problem and our solution

22.8%

of homes are high-rise: **no roof** for solar

DOSM Census 2020

18 : 1

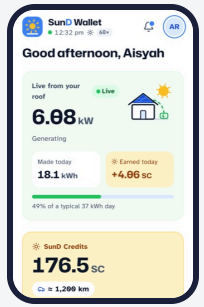
EVs per public charging bay: **no clean charging**

122,286 battery EVs ÷ 6,904 public bays (Kenanga, Business Today 2026)

0 kWh

of Elmina's CREAM solar reaches **condo residents**

CREAM roofs sell their power to SDP's Innovation Park (SDP, 2025)



CORE 1

SunD Wallet

Township solar → credits for residents



CORE 2

SunD Bay

Solar canopy + battery EV charging

Around it: SunD Walk · SunD Cool · eBikes · balcony kits (2027)

→ This week we tested: **transmission cost · upfront cost · e-Panel · Saver & Pass · business model**

Source: original pitch deck, slides 2–4.

TRANSMISSION COST · 1/2

Solar is worth more used where it's made

We compared the same 1 kWh of solar, used two ways

A · Use it on site

Panels feed the building's own meter (Solar ATAP / SELCO). Each kWh saves the building's **full TNB rate**.

B · Send it through TNB's lines (CREAM)

Power goes to a buyer ≤ 5 km away. The buyer saves only the **energy part** of the bill, and the seller pays a **9 sen toll** per kWh.

VALUE OF 1 KWH, SEN	A · ON SITE	B · THROUGH TNB	COMPARE	VALUE LOST
Mall medium-voltage meter	**33.5** 29.83 energy + 3.67 AFA	**24.5** 29.83 energy + 3.67 AFA − 9 toll		**−27%** (33.5 − 24.5) ÷ 33.5
SunD Bay / clubhouse low-voltage meter	**54.4** 27.03 energy + 8.83 capacity + 14.82 network + 3.67 AFA	**21.7** 27.03 energy + 3.67 AFA − 9 toll		**−60%** (54.4 − 21.7) ÷ 54.4
Landed home domestic meter	**44.4** 27.03 energy + 4.55 capacity + 12.85 network	**18.0** 27.03 energy − 9 toll		**−59%** (44.4 − 18.0) ÷ 44.4

TNB tariff, Jul 2025 energy, capacity and network charges per kWh · TNB / EC AFA = monthly fuel surcharge, +3.67 sen in Sep 2026 · Energy Commission 9 sen = CREAM access charge (was 15 sen) · CREAM only allows landed rooftops, not mall canopies

Source: TNB tariff schedule (1 Jul 2025) via myenergystats.st.gov.my; AFA Sep 2026 (paultan.org); Sheet Research list R-16, R-76, R-77, R-89, R-97.

sourced assumption design choice model

Check: SDP's own CREAM programme never pays back

450 Elmina homes host rooftop solar; SDP sends the power through TNB's lines to its Innovation Park

What goes in

INPUT	VALUE	SOURCE
Solar installed	450 homes × 10 kWp = **4,500 kWp**	SDP report 2024
Solar made	3.6 kWh per kWp a day → 4,500 × 3.6 × 365 = **5.91M kWh a year**	PVGIS
Price Innovation Park pays	**RM 0.27** per kWh: the most a CREAM kWh is worth (energy part only, p3); real price is private	estimate
Toll (access charge)	**RM 0.09** per kWh sent	Energy Commission
Roof lease	rent SDP pays each homeowner: **RM 50** a month	placeholder
Upkeep	cleaning and repairs: **RM 50** per kWp a year (market RM 75–90, so likely worse)	placeholder
Build cost	**RM 3,500** per kWp installed (quotes RM 3,000–4,000)	installer quotes

What comes out, each year

Revenue	5.91M kWh × RM 0.27	**RM 1.60M**
– Toll	5.91M kWh × RM 0.09	**–RM 0.53M**
– Roof lease	450 homes × RM 50 × 12	**–RM 0.27M**
– Upkeep	4,500 kWp × RM 50	**–RM 0.23M**
= Profit a year	1.60 – 0.53 – 0.27 – 0.23	**RM 0.57M**

BUILD COST
RM 15.75M
4,500 kWp × RM 3,500

PAYBACK
27.7 yrs
15.75 ÷ 0.57 · longer than the 25-yr panel life

NPV model
–RM 9.67M
value today of 25 yrs of profit at 8% a year, minus the build cost

→ Our design: **use solar where it's made → battery for the evening → TNB grid only tops up**

Source: Sheet CREAM tab and Inputs R1–R7 (8% discount rate = SDP's cost of money, 25 yrs); Research list R-01, R-02, R-13, R-78, R-89. SDP has since stopped the programme.

sourced assumption design choice model

UPFRONT COST · 1/2

One 30-lot SunD Bay costs RM 1.76M to build

30 lots · one charger per lot (6 DC + 24 AC) · 75 kWp canopy (30 lots × 2.5 kWp) · 180 kWh battery for the evening

ITEM	QTY × UNIT COST	RM	SHARE	WHERE THE UNIT COST COMES FROM
DC chargers 60 kW	6 × RM 70,000	**420,000**	24%	RM 60–80k installed Trexon 2026
AC chargers 22 kW	24 × RM 11,500	**276,000**	16%	RM 8–15k installed Trexon 2026
Solar canopy	75 kWp × RM 4,250	**318,750**	18%	middle of RM 3,500–5,000 per kWp Cobler, Progressture
Battery	180 kWh × RM 1,750	**315,000**	18%	middle of RM 1,500–2,000 per kWh assumption
Grid connection + substation	one-off	**300,000**	17%	TNB RM 45/kW + substation; RM 0 if mall has spare capacity assumption
Civil works, metering, design	10% × RM 1,329,750 equipment	**132,975**	8%	assumption
Total, one bay		**1,762,725**	**100%**	

REPLACEMENTS LATER	CALCULATION	RM	WHERE IT COMES FROM
Year 9 · DC charger power modules	25% × RM 420,000	**105,000**	modules wear out in 8–10 yrs assumption
Year 12 · inverters	75 kWp × RM 350	**26,250**	RM 270–400 per kWp installed R-78
Year 16 · battery	60% × RM 315,000	**189,000**	battery prices keep falling assumption

Source: Sheet Inputs S4–S8, C1–C10, L5, M1–M4; Summary (build cost RM 1,762,725); Research list R-13, R-14, R-15, R-77, R-78.

sourced assumption design choice model

UPFRONT COST · 2/2

Who pays? Prepaid buyers fund the solar, SDP the rest

SunD's own business at the start: 2 SunD Bays + solar on the mall, clubhouse and hall roofs

WHAT WE BUILD AT THE START	CALCULATION	RM
2 SunD Bays	2 × RM 1.76M (p5)	**3.53M**
Roof solar on mall, clubhouse, hall	714 kWp × RM 3,500 per kWp	**2.50M**
Total build		**6.02M**

Who pays for it

Prepaid buyers **RM 3.02M** | SDP **RM 3.00M**

Prepaid buyers pay for the **solar panels**

roof panels RM 2.50M + bay canopy panels 150 kWp × RM 3,500 = RM 0.53M

SDP pays for **everything else**: chargers, batteries, canopy frames, grid link

RM 6.02M − RM 3.02M = RM 3.00M · plus RM 0.12M to sell the packs (5% × RM 2.50M) = **RM 3.13M SDP cash at the start**

Buyers prepay ringgit and get SunD credits worth 30% more (p7, p8); SDP keeps the panels and the power for 25 yrs

design choice

What helps with the cost

Tax allowance (GITA) LHDN / MGTC

Not cash: SDP deducts 60% of what it spends on solar (100% for batteries) from taxable profit. Saves about **14.4%** of the solar cost.

60% × 24% company tax = 14.4% · for EV chargers it drops from 100% to 60% in 2027 (Budget 2027)

Cash grants MIDA, SEDA, Selangor

None for a commercial solar car park or chargers.

Green loans RHB, BNM

From **4.5%** a year (RHB); Bank Negara fund up to RM 10M at ≤ 5%.

Source: Sheet Precinct strategies (option 5: SDP cash at year 0 RM 3,125,400; option 1: RM 6,024,450), Inputs G3, G15, G21; Research list R-19 (GITA, Budget 2027), R-38, R-39. Year-2 canopy expansion and township items (home roofs, SunD Walk) not included.

sourced assumption design choice model

FEATURE · SUND E-PANEL

SunD e-Panel

REJECTED

Idea: residents buy a solar panel on an SDP roof and get SunD credits back

VERSION TESTED	BUYER PAYS	BUYER GETS	MONEY BACK IN	RESULT
v1	RM 2,608 a panel RM 1,925 panel + RM 683 share of a mall battery	30 sen credit per kWh, 20 yrs design choice 0.55 kWp × 1,300 kWh × 30 sen = RM 214 a year	**12 yrs** 2,608 ÷ 214	~5% a year, below EPF (~6%): nobody buys
v2	RM 1,925 a panel 0.55 kWp × RM 3,500 per kWp	RM 322 credit a year, 10 yrs design choice 715 kWh × 45 sen	**6 yrs** 1,925 ÷ 322	10-yr wait, reads like an investment
v3	RM 385 for 0.2 panel 0.2 × RM 1,925	RM 500 of credits at once design choice	**day one** 500 ÷ 385 = 1.30, i.e. +30%	works, but “own a panel” is still an investment risk

One panel = 0.55 kWp (550 W) · 1,300 kWh per kWp a year (conservative; Malaysia 1,350–1,450) · RM 3,500 per kWp installed (R-13)

Investment law

Paying for a panel to get a benefit from SDP’s effort counts as an **interest scheme**: register it, or up to 10 yrs jail / RM 50M fine Interest Schemes Act 2016

Slow for buyers

v1–v2 take 6–12 yrs to pay back; a home’s own roof solar pays back in **5.2 yrs**

5 kW system RM 19,000 ÷ RM 3,672 saved a year (R-85)

Hard to explain

Panel ownership, sell-back prices and 20-yr terms: too complex for a resident buying a discount

→ Kept: the prepay idea (pay RM 1, get RM 1.30 of credits) as **Saver top-ups**, with no panel attached (p8)

Source: research doc tabs 1–3; Sheet Design review 1, 2, 18; Inputs G1–G21; Research list R-13, R-79, R-85; Credit plan (“adopt a panel” dropped 10 Nov).

sourced assumption design choice model

FEATURES · SAVER CREDITS + SUND PASS

Saver top-ups chosen; SunD Pass works for public drivers

Saver top-ups · top up more, get more design choice

PAY	GET IN CREDITS	BONUS	SAME AS % OFF 1 − pay ÷ get
RM 50	RM 55	+10%	**9% off**
RM 100	RM 115	+15%	**13% off**
RM 200	RM 240	+20%	**17% off**
RM 300	RM 375	+25%	**20% off**
RM 500	RM 650	+30%	**23% off**

MARKET TODAY operator price lists	PRICE	% OFF
JomCharge Elite	RM 9.90 a month	10%
Gentari GO	RM 350–699 a year	15–30%
Shell Recharge Gold	RM 835 a year	40%
Our old Saver: RM 20 → 30, free original deck	cut: beat every paid plan	33%

SunD Pass · one monthly price design choice

RM 79 residents · **RM 129** public · up to 100 kWh of charging + eBike rides + mall parking

WHAT A SUBSCRIBER PAYS VS GETS, A MONTH	RESIDENTS	PUBLIC
Pays for the Pass	**RM 79**	**RM 129**
Same charging at the normal price 85 kWh used × RM 0.90 residents / RM 1.40 public	**RM 76.50**	**RM 119**
+ eBike rides and parking included assumed use, at list price	**RM 40**	**RM 40**
Value the subscriber gets	**RM 116.50**	**RM 159**

Source: Sheet Credit plan, Inputs D6, D7, D9, D12, G21, SunD Pass tab SP1–SP14, Design review 4; Research list R-68, R-82 (operator price lists). 85 kWh = 85% of the 100 kWh used (assumption).

sourced assumption design choice model

BUSINESS MODEL · 1/2

6 ways to run it: who pays is the difference

Same 2 SunD Bays + roof solar in every option · NPV = value today of 25 yrs of cash at 8% a year · SDP cash = what SDP puts in at the start

#	WHO PAYS	NPV TO SDP model	SDP CASH model	VERDICT
1	SDP pays for everything; no customer product	RM 1.52M	RM 6.02M	No money loop
2	SDP pays; renters get free credit slices (original deck)	RM 1.00M	RM 6.02M	Gives credits away
3	Buyers get tradeable tokens (NFT) + a cash share	−RM 0.59M	RM 5.65M	Rejected
4	A partner owns the bays; SDP pays the roof solar	RM 0.94M	RM 2.50M	Fallback
5	**SDP builds the bays; prepaid buyers fund the solar**	RM 2.26M	RM 3.13M	Chosen
6	A partner owns the bays; prepaid buyers fund the solar	RM 1.38M	RM 0.12M	Fallback

Option 1: SDP pays the full RM 6.02M build (p6) · option 5: RM 3.13M (p6) · "prepaid buyers" = Saver top-ups: pay RM 1, get RM 1.30 of credits (p8)

Why not NFT, tokens or an investment product?

The law

A tradeable token is a **security**: SC approval, RM 2,000 cap per buyer Securities Commission. Selling for a return is an **interest scheme**: register, or up to 10 yrs jail / RM 50M Interest Schemes Act

The money

Tokens lose **RM 0.59M** (option 3). The same roofs sold as an investment earn RM 0.14–0.30M vs **RM 1.20M** as prepaid credits model

The buyers

Token buyers wait **9 yrs** to get their money back model ; a 6% cash return only matches EPF (~6%)

Source: Sheet Precinct strategies, Why this model, Public investment tab, Business models (model A); Research list R-79, R-93; Inputs G21. Option 5 was modelled as the e-Panel pack; the same RM 1 → RM 1.30 prepay now runs as Saver top-ups.

sourced assumption design choice model

BUSINESS MODEL · 2/2

Chosen: SDP builds the bays, prepaid buyers fund the solar

Option 5 (p9) · 2 SunD Bays + 714 kWp roof solar (+ canopy expansion in year 2) · 25 yrs · before tax model

NPV TO SDP
RM 2.26M
value today of 25 yrs of cash, at 8% a year

IRR
12%
yearly return on the money put in

PAYBACK
9 yrs
until cash in covers cash out

SDP CASH AT THE START
RM 3.13M
of the RM 6.02M build (p6)

Why this one

1. **Use power where it's made** · no 9 sen toll (p3)
2. **Pay people in credits, not cash** · RM 1 of credit costs SDP about RM 0.65
   model
   most credits are spent on SDP's own services; e.g. RM 1 of charging only costs SDP the electricity behind it
3. **Sell a discount, not an investment** · no scheme to register (p9)

If nobody prepays, SDP pays for the solar itself (option 1): NPV still **RM 1.52M**.

How sure

	BEAR	BASE	BULL
Charging sessions per lot a day assumption	1.3	1.6	2.0
Running cost per bay a day assumption	RM 180	RM 180	RM 120
Grid connection per bay assumption	RM 450k	RM 300k	RM 0
Years buyers take to spend credits assumption	1	2	4
NPV model	**RM 0.69M**	**RM 2.26M**	**RM 4.82M**
Payback model	10.6 yrs	9.0 yrs	7.0 yrs

→ Need from SDP: **car-park traffic · transformer capacity · mall load data** · From regulators: **SSM view · BNM notice · SEDA registration**

Source: Sheet Summary (NPV RM 2,256,654, IRR 12.0%, payback 9.0), Precinct strategies, Why this model, Scenarios tab, USP money loop (credit cost 0.65).

sourced assumption design choice model